

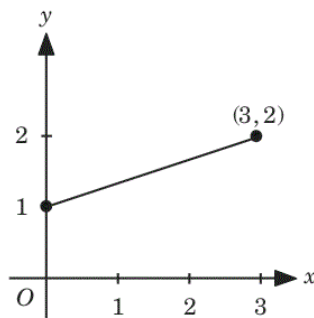
Note: DO NOT USE A CALCULATOR. Calculators are not allowed on the Placement Exam!

1. If $a = 2$ and $b = -3$, find the value of $\frac{(a - b)^2 + b}{(b - 2a)^2 + a}$.

2. Simplify

$$2(x + 3xy + y^2) - (-x^2 + 6xy + 4y)$$

3. What is the equation of the line segment given in the following figure.

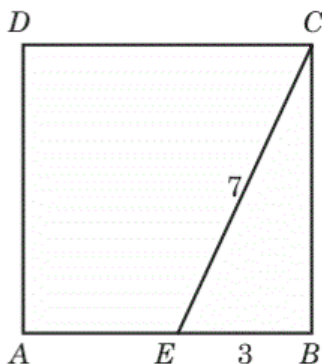


4. Solve the system of equations

$$\begin{cases} 3x + y = 1 \\ x - 3y = 17 \end{cases}$$

5. If $f(x) = x^2 + 2x + 3$, find $f(a - 1)$. Simplify your answer.

6. What is the area of the square $ABCD$ given in the figure below.



7. For what value of t does

$$\frac{2t - 1}{t + 3} = -2.$$

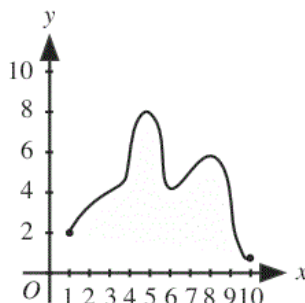
8. Express the following with all positive exponents.

$$(27a^{-3}b^6c^3)^{1/3}$$

9. Simplify

$$\frac{x+1}{x(x-1)} - \frac{1}{2(x-1)}$$

10. The graph of a function $y = f(x)$ is shown in the figure below. For how many values of x is $f(x) = 3$?



11. A ball is kicked into the air at time $t = 0$. While the ball is in the air, its position above the ground after t seconds is given by $h(t) = -16t^2 + 40t$ feet. How many seconds after the ball is kicked will it hit the ground?

12. Simplify

$$\frac{x^{4b+1}}{x^{2-b}}.$$

13. There are 35 men in a room. If 4 of every 5 occupants of the room are women, how many women are in the room?

14. If $f(x) = x^2 - 3x$ and $g(x) = \sqrt{x} - 1$, find $f(g(x))$.

15. Solve for x if

$$8^{2-x} = 4^{3x}.$$

16. Find the solutions to the equation

$$x^2 - 5x - 2 = 0$$

17. The solution set to the inequalities $0 \leq x \leq 2$, $0 \leq y \leq 1$ and $y \leq \frac{-1}{2}x + \frac{3}{2}$ form a polygonal region. How many sides does the polygon have?

18. Plot the equation

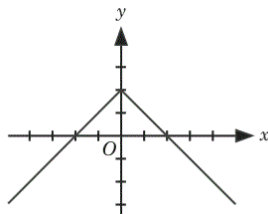
$$x^2 - 6x + y^2 + 2y + 6 = 0.$$

19. Find the solution set to the inequality $|x - 4| > 3$.

20. Simplify

$$\frac{\frac{21-7x}{x+3}}{\frac{x^2-3x}{2x+3}}$$

21. The plot of $h(x)$ is given the figure below. Sketch a plot of $h(x + 2)$.



22. Write a system of equations that describes the following situation. You do not need to solve.

A collection of 15 coins, consisting of only nickels and quarters, has a total value of \$2.75. Let x be the number of nickels (5 cents each), and y be the number of quarters (25 cents each).

23. Find w if

$$\ln w = \frac{1}{5} \ln x - \ln y.$$

24. Find the solution set to the inequality

$$x^2 - 2x < 8.$$

25. suppose that $b \neq 3$. Solve for x if

$$x + a = \frac{b}{3}x.$$

26. Sketch a plot of the function

$$f(x) = \left(\frac{1}{2}\right)^x + 3.$$

27. Simplify

$$\sqrt[4]{3} \cdot \sqrt[5]{3}$$

28. Simplify

$$\sqrt{4x^4 + 36x^2 + 81}$$

29. Find an equation for the horizontal asymptote of

$$y = 3 - \frac{x + b}{x - c}.$$

30. Solve

$$8 = \sqrt{2x - 3} + 3$$